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BSED SCIENCE 2C

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Lecture Quiz No. 2

BACTERIAL STRUCTURES and FUNCTIONS

1. Discuss the structure of a typical bacterial cell and explain how each major component contributes to bacterial survival and metabolism.

In your answer, refer to the cell wall, plasma membrane, cytoplasm, nucleoid, ribosomes, and any additional structures where relevant.

- Cell wall: Gives the bacterium shape and strength. It protects the cell from bursting if the environment changes (like too much water).
- Plasma (cell) membrane: A thin layer under the cell wall that controls what enters and leaves the cell, like nutrients and waste.
- Cytoplasm: A jelly-like substance inside the cell where chemical reactions happen. Think of it as the “workshop” of the cell.
- Nucleoid: The region where bacterial DNA is stored. It controls the cell’s growth, reproduction, and activities.
- Ribosomes: Small structures that make proteins, which the cell needs to grow and function.

2. Explain the differences between Gram-positive and Gram-negative bacteria, focusing on cell wall structure and functional significance.

- Gram-positive:
  - Thick peptidoglycan cell wall (many layers)
  - No outer membrane
  - Stains purple in Gram stain
  - Strong and protects the cell well, but doesn’t block antibiotics as much

- Gram-negative:
  - Thin peptidoglycan cell wall (few layers)
  - Has an outer membrane with lipopolysaccharides
  - Stains pink/red in Gram stain
  - Outer membrane protects them from many antibiotics and harsh chemicals

3. Describe the role of bacterial surface structures in attachment, movement, and virulence.

- **Pili or fimbriae** are hair-like projections that allow bacteria to **attach to surfaces or other cells.**
- **Flagella** are tail-like structures that help bacteria **move toward food or away from danger.**
- The **capsule** is a slimy layer that **protects bacteria from the immune system and harsh conditions,** allowing them to survive and infect hosts.

4. Analyze the importance of genetic structures in bacteria.

Discuss the functions of the nucleoid and plasmids, and explain how genetic exchange mechanisms contribute to bacterial evolution and antibiotic resistance.

- Functions of the nucleoid:
  - The nucleoid is the region in the bacterial cell where the **main DNA is stored**. It contains the essential genes that control **growth, metabolism, and reproduction** of the bacterium.

- Functions of plasmids:

  - Plasmids are **small extra pieces of DNA** separate from the main DNA. They can carry genes that give bacteria **special advantages**, such as **resistance to antibiotics** or the ability to use new nutrients.

- Genetic exchange and its significance:

  - **Conjugation:** Transfer of plasmids through pili to another bacterium.
  - Transformation: Picking up DNA from the surrounding environment.
  - **Transduction:** DNA is transferred by viruses from one bacterium to another

5. Evaluate the adaptive significance of specialized bacterial structures such as endospores and capsules.

Explain how these structures enable bacteria to survive extreme/hostile/ harsh environmental conditions and contribute to their persistence in nature and clinical settings.

- Endospores:

➢ Endospores are **tough**, **dormant structures** that some bacteria form when conditions are harsh. They protect the bacterium from **heat**, **drying**, **chemicals**, and **other extreme conditions.** This allows bacteria to **survive for long periods** until the environment becomes favorable again.

- Capsules:

➢ Capsules are **slimy layers** surrounding the bacterial cell. They **protect bacteria from the immune system**, help them stick to surfaces, and allow them to **survive in harsh environments**. Capsules make bacteria **harder to kill in nature and in clinical settings**, which is why some infections are difficult to treat.